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Lithium Ionic Conductor Thio-LISICON

The $\text{Li}_2\text{S-GeS}_2\text{-P}_2\text{S}_5$ System

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The new crystalline material family, lithium superionic conductor (thio-LISICON), was found in the $\text{Li}_2\text{S-GeS}_2\text{-P}_2\text{S}_5$ system. The solid solution member $x = 0.75$ in $\text{Li}_{4-x}\text{Ge}_{1-x}\text{P}_x\text{S}_4$ shows the highest conductivity of $2.2 \times 10^{-3} \text{ S cm}^{-1}$ at 25°C of any sintered ceramic, together with negligible electronic conductivity, high electrochemical stability, no reaction with lithium metal, and no phase transition up to 500°C . Its material design concepts of changing constituent ions with various ionic radii, valence, and polarizability are described.

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Lithium ionic conductors exhibit high lithium ionic diffusion in one of their ionic sublattices, the mobile ion sublattice, at temperatures well below the melting points.¹ It is of great interest to understand their fast ionic transport in “solid” materials, which is widely recognized, although still a relatively rare phenomenon, it is also of technological importance for developing solid electrolytes for advanced lithium batteries.² Generally speaking, crystalline ionic conductors have a number of advantages in conductivity and stability over glassy or polymer systems, *i.e.*, crystalline copper and silver ionic conductors show extremely high conductivities even at room temperature,³⁻⁵ but no convincing examples of such concept have been found in the lithium conductors.² Material design of crystalline ionic conductors are based on certain structural criteria:⁶ (i) mobile ions should have a suitable size for conduction pathways in the lattice, (ii) there should be disorder in a mobile ion sublattice, and (iii) highly polarizable mobile ions and anion sublattices are preferable. Lithium superionic conductor (LISICON) [$\text{Li}_{14}\text{Zn}(\text{GeO}_4)_4$]⁷ is an example of a material which was designed using the above criteria. A wide variety of materials with the same type of framework have been synthesized; a wide range of the solid solutions formed by aliovalent substitutions introduced interstitial lithium ions or vacancies and led to high ionic conductivities at elevated temperatures.¹ However, the conductivity values at room temperature ($\approx 10^{-6} \text{ S cm}^{-1}$ for $\text{Li}_{3.6}\text{Si}_{0.6}\text{P}_{0.4}\text{O}_4$,^{8,9} for example) are still lower than those of copper and silver ionic conductors.³⁻⁵ In order to modify these materials to obtain high ionic conductivity, sulfides have a number of advantage over oxides for constructing ionic conductors according to the above criteria of material design; *i.e.*, larger ionic radii and more polarizable character of sulfide ions may improve the mobility of the conducting species. This has been noted in the sulfide glasses; high conductivity values of 10^{-4} to $10^{-3} \text{ S cm}^{-1}$ at room temperature were obtained in the $\text{Li}_2\text{S-SiS}_2\text{-LiI}^{10}$ and $\text{Li}_2\text{S-SiS}_2\text{-Li}_3\text{PO}_4$ ¹¹ glasses. Generally speaking, crystalline materials should have higher conductivity than the corresponding glasses, if their crystal structures have been well designed for high ionic conduction. For example, the highest ionic conductive solids, $\text{Rb}_4\text{Cu}_{16}\text{I}_7\text{Cl}_{13}$ ^{3,4} and RbAg_4I_5 ,⁵ have higher ionic conductivity than any copper and silver conducting glasses. However, this is not true for crystalline lithium ionic conductors whose ionic conductivities and electrochemical decomposition potentials are usually lower than glassy materials. Therefore, almost no effort has been made into material developments concerning crystalline lithium ionic conductors during the last decade, while a great improvement on glassy lithium ionic conductors¹²⁻¹⁴ has led to the developmental research on all solid-state lithium battery.^{2,15-17} Therefore, we have initiated a new materials search on crystalline lithium ion conductors. In the

current study, material based on the recently discovered lithium germanium sulfides, Li_4GeS_4 ,¹⁸ and a new thio-LISICON, $\text{Li}_{4-x}\text{Ge}_{1-x}\text{P}_x\text{S}_4$, was found with the substitution, $\text{Ge}^{4+} + \text{Li}^+ \leftrightarrow \text{P}^{5+}$ in the $\text{Li}_2\text{S-GeS}_2\text{-P}_2\text{S}_5$ system.

Experimental

Starting materials, Li_2S , GeS_2 , and P_2S_5 (Pure Chemical Ind. Ltd., >99.9% purity) were weighed, mixed in appropriate molar ratios in an argon filled glove box. The mixture was sealed in a carbon-coated quartz tube and heated to the reaction temperature of 700°C for 8 h. After the reaction, the tube was slowly cooled to room temperature. The reaction products were characterized by X-ray diffraction (XRD) study using $\text{Cu K}\alpha$ radiation (Rigaku Rotaflex 12 kW rotating anode diffractometer). The diffraction data were collected at each 0.02° step width over a 2θ range from 20 to 110° . The lattice parameters were determined by Rietveld analysis¹⁹ using the computer program RIETAN-2000.

Ionic conductivity was measured using an ac impedance method in a nitrogen gas flow over a $25 \leq t \leq 300^\circ\text{C}$ temperature range with an applied frequency range of 0.1 Hz to 1 MHz using Solatron 1260 frequency response analyzer. Gold paste used blocking electrodes were painted onto both sides of a pellet (8 mm diam and $2\text{--}3 \text{ mm}$ thick) and the pellet was then heated under vacuum at 150°C for 3 h. The conductivity values were determined from the impedance plots of the data. The $\text{Li/Li}_{3.25}\text{Ge}_{0.25}\text{P}_{0.75}\text{S}_4/\text{Au}$ cell was measured by cyclic voltammogram (CV) using a reference lithium electrode with the scan rate of 1 mV/s and the scan range between -0.5 and 5 V using a solatron 1287 electrochemical interface. Electronic conductivity was measured by the Hebb-Wagner polarization technique²⁰ using a $(-)\text{Li/Li}_{3.25}\text{Ge}_{0.25}\text{P}_{0.75}\text{S}_4/\text{Au}(+)$ at 25°C .

Results and Discussion

Synthesis and structures.—Figure 1 shows the XRD patterns of the $\text{Li}_{4-x}\text{Ge}_{1-x}\text{P}_x\text{S}_4$ synthesized in the present study. The structures of the end members, Li_4GeS_4 ¹⁸ and Li_3PS_4 ,²¹ were confirmed by the Rietveld analysis to have similar structure to $\gamma\text{-Li}_3\text{PO}_4$. The diffraction data of the end members, Li_4GeS_4 and Li_3PS_4 , were refined using the structure models of $\text{Li}_{3.5}\text{Zn}_{0.25}\text{GeO}_4$,²² and Li_3PS_4 ,^{21,23} respectively. The refined cell parameters of $a = 14.0562(3)\text{\AA}$, $b = 7.7462(2)\text{\AA}$, and $c = 6.14579(15)\text{\AA}$; and $a = 13.1320(11)\text{\AA}$, $b = 7.7402(7)\text{\AA}$, and $c = 6.1230(4)\text{\AA}$, for Li_4GeS_4 and Li_3PS_4 , respectively, were consistent with those reported previously.^{18,21} The thio-LISICON has a similar structure to that of the LISICON (Fig. 2).

The diffraction pattern showed a continuous shift in the diffraction lines from Li_4GeS_4 to Li_3PS_4 , which suggests the formation of a solid solution in the whole composition range. Careful analysis of the diffraction patterns, however, indicated that the system is divided into three composition regions depending on the appearance of superlattice reflections. The patterns in region I ($0 < x \leq 0.6$) were indexed assuming a monoclinic superlattice cell of

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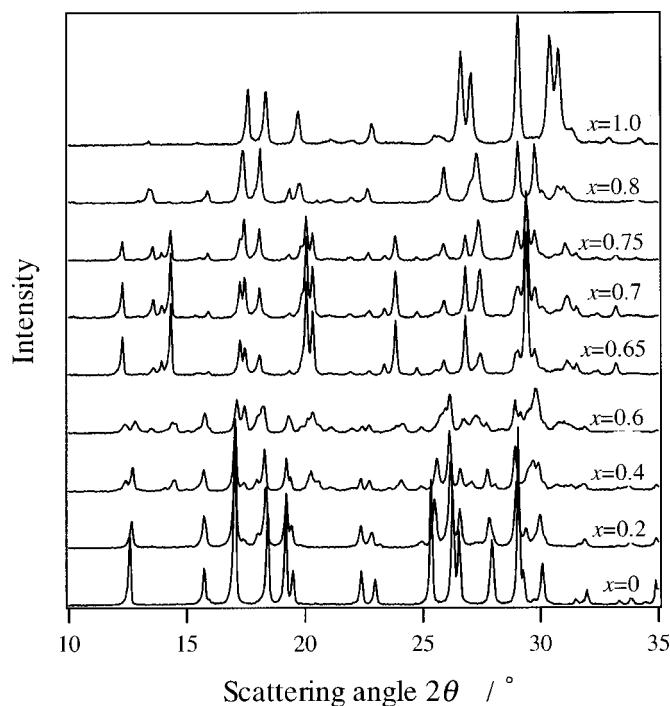


Figure 1. XRD patterns of $\text{Li}_{4-x}\text{Ge}_{1-x}\text{P}_x\text{S}_4$.

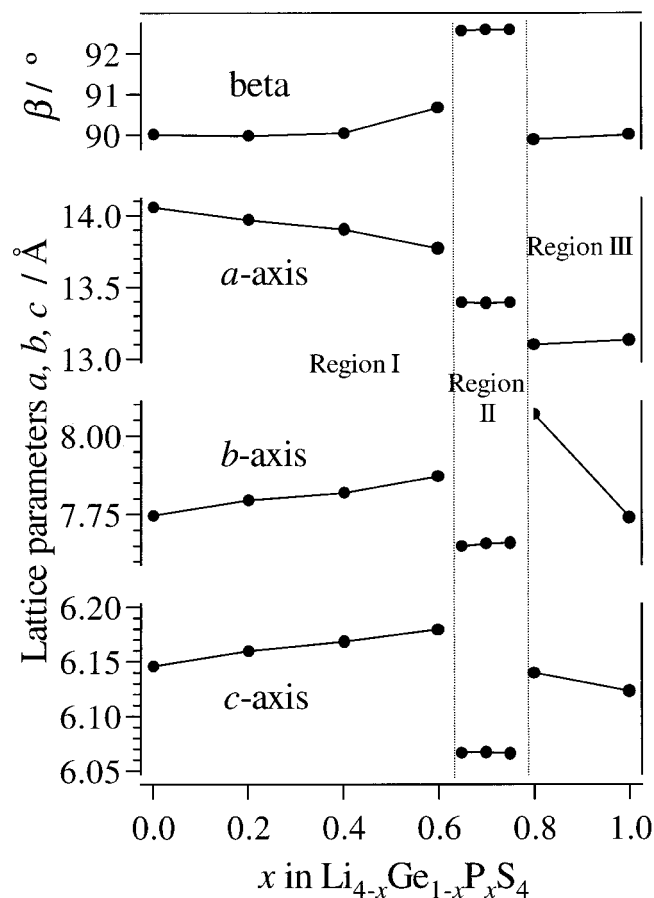


Figure 3. Composition dependence of the lattice parameters in $\text{Li}_{4-x}\text{Ge}_{1-x}\text{P}_x\text{S}_4$ determined by XRD measurements. The lattice parameters plotted in the figure are based on the parent lattice of the LISICON with the $a \times b \times c$ cells.

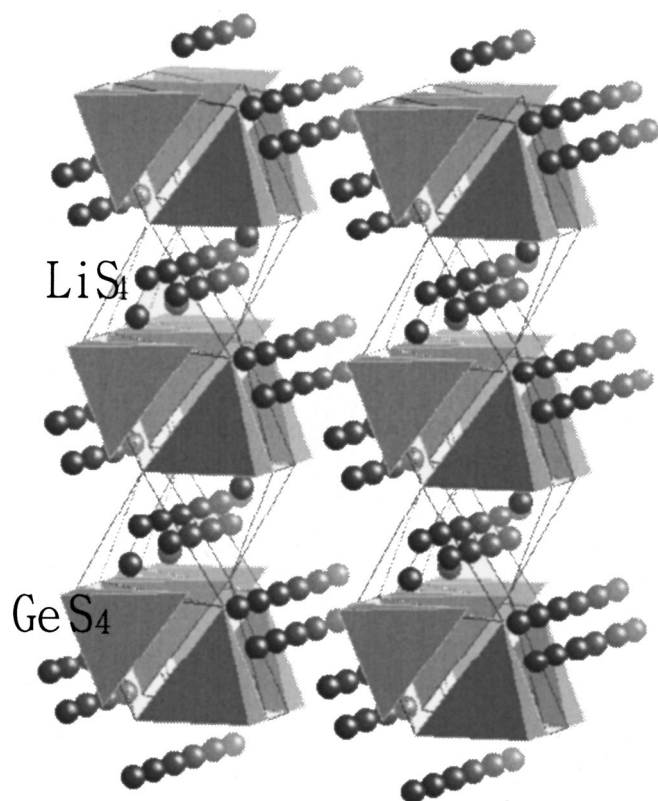


Figure 2. The structure of the thio-LISICON. The structure drawn in the figure is based on the parent structure of Li_4GeS_4 . The three-dimensional framework structure is composed of GeS_4 tetrahedra. Possible lithium sites are indicated in the figure.

$a \times 3b \times 2c$ which is related to the parent lattice of the orthorhombic Li_4GeS_4 . However, the superlattice reflections in region II ($0.6 < x < 0.8$) were indexed by a different monoclinic cell with $a \times 3b \times 3c$, and the sample at $x = 0.8$ in region III ($0.8 \leq x < 1.0$) showed the monoclinic $a \times 3b \times 2c$ cell. The three regions of the monoclinic superstructures correspond to different types of cation ordering.

The lattice parameters of the solid solutions were refined using the superlattice structures suitable for each region with space group $P2_1/m$. Figure 3 shows the composition dependence of the lattice parameters in $\text{Li}_{4-x}\text{Ge}_{1-x}\text{P}_x\text{S}_4$ determined by XRD measurements. The lattice parameters plotted in Fig. 3 are based on the parent lattice of the LISICON with the $a \times b \times c$ cells. Table I summa-

Table I. Lattice parameters in $\text{Li}_{4-x}\text{Ge}_{1-x}\text{P}_x\text{S}_4$.

x	$a/\text{\AA}$	$b/\text{\AA}$	$c/\text{\AA}$	$\beta/^\circ$
0.0	14.0562(3)	7.7462(2)	6.14579(15)	90.000
0.2	13.9668(9)	23.3842(15)	12.3196(7)	89.967(19)
0.4	13.9016(13)	23.454(2)	12.3367(9)	90.043(19)
0.6	13.769(4)	23.614(5)	12.359(2)	90.66(3)
0.65	13.393(3)	22.947(6)	18.2009(15)	92.562(14)
0.7	13.387(3)	22.968(5)	18.2015(16)	92.594(14)
0.75	13.395(3)	22.979(5)	18.1981(18)	92.591(14)
0.8	13.0981(14)	24.201(2)	12.2793(7)	89.887(18)
1.0	13.1320(11)	7.7402(7)	6.1230(4)	90.000

izes the lattice parameter values of $\text{Li}_{4-x}\text{Ge}_{1-x}\text{P}_x\text{S}_4$. Detailed neutron studies are needed to clarify the structural details of the superstructures.

Ionic and electronic conductivities.—Typical behavior of the impedance plots could be described by the combination of two components, a semicircle in the high frequency range and a spike in the low frequency region. The former is attributed to the bulk and grain boundary contribution, and the latter to the electrode contribution.²⁴ The conductivity values were obtained by a least-squares fitting based on a parallel combination of resistance and capacitance with the Warburg impedance connected in series. This type of behavior is characteristic of a pure ionic conductor, and was confirmed by electronic conductivity measurements (see below). No differences in the conductivities were observed between the cooling and heating process during the measurements. The temperature dependences of the conductivity in $\text{Li}_{4-x}\text{Ge}_{1-x}\text{P}_x\text{S}_4$ are shown in Fig. 4a. Plots of the composition vs. the conductivity and the activation energy are shown in Fig. 4b. The conductivity increased continuously as the P content increased from $x = 0.0$ to $x = 0.75$ in $\text{Li}_{4-x}\text{Ge}_{1-x}\text{P}_x\text{S}_4$, and the highest conductivity value of $2.17(6) \times 10^{-3} \text{ S cm}^{-1}$ at 25°C was obtained. At the same time, the activation energy decreased with x to the lowest value of 20 kJ mol^{-1} at $x = 0.75$. In aliovalent solid solution, ions are substituted by other ions of different charge, which consequently causes additional changes involving creation of vacancies or interstitials (ionic compensation), or electrons or holes (electronic compensation) to preserve electroneutrality. It is reasonable to assume the former mechanism because no transition elements exist as a constituent of the thio-LISICON, which likely forms pure ionic conduction as a result. In order to confirm this, the electronic conductivity was measured by the Hebb-Wagner polarization technique²⁰ using a $(-)\text{Li}/\text{Li}_{3.25}\text{Ge}_{0.25}\text{P}_{0.75}\text{S}_4/\text{Au}(+)$ cell at 25°C . Figure 5 shows the Hebb-Wagner polarization curve. The total electronic conductivity ($\sigma_e + \sigma_h$) at the irreversible $\text{Au}-\text{Li}_{3.25}\text{Ge}_{0.25}\text{P}_{0.75}\text{S}_4$ interface of the asymmetric cell is given by²⁰

$$\sigma_e + \sigma_h = (L/S)(d|I_{e,h}|/dE) \quad [1]$$

where L represents the electrolyte thickness, S the electrolyte-electrode surface area, $|I_{e,h}|$ the absolute value of the total electronic current, and E the polarization voltage. The total electronic conductivity ($\sigma_e + \sigma_h$) at the irreversible $\text{Au}-\text{Li}_{3.25}\text{Ge}_{0.25}\text{P}_{0.75}\text{S}_4$ interface of the asymmetric cell was calculated by the linear fitting between 1.0 and 1.5 V in Fig. 5, and was determined to be $2.6 \times 10^{-9} \text{ S cm}^{-1}$. For the reversible $\text{Li}-\text{Li}_{3.25}\text{Ge}_{0.25}\text{P}_{0.75}\text{S}_4$ interface, the conductivity of electrons and holes can be determined separately from the slope and the axial intersection using the following equation²⁰

$$|I_{e,h}|(e^{-EF/RT} - 1) = (RTS/FL)(\sigma_e e^{-EF/RT} + \sigma_h) \quad [2]$$

where T represents absolute temperature, R gas constant, and F Faraday constant. The conductivity of electrons, σ_e , and holes, σ_h , at the reversible $\text{Li}_{3.25}\text{Ge}_{0.25}\text{P}_{0.75}\text{S}_4$ -Li interface was calculated from the $I_{e,h} \text{ FL}/RTS(e^{-EF/RT} - 1)$ vs. $e^{-EF/RT}$ plots and was determined to be 1.5×10^{-7} and $4.5 \times 10^{-19} \text{ S cm}^{-1}$, respectively. Considering the total conductivity measured by the ac impedance method, the electronic contribution to the total conductivity is negligible. The new thio-LISICON is a pure ionic conductor and, consequently, the partial substitution of P^{5+} for Ge^{4+} introduced lithium vacancy and increased the conductivities. The conductivity value of $2.17(6) \times 10^{-3} \text{ S cm}^{-1}$ at 25°C is much higher than those of $\text{Li}_{14}\text{Zn}(\text{GeO}_4)_4$ ⁷ and $\text{Li}_{3.6}\text{Si}_{0.6}\text{P}_{0.4}\text{O}_4$ ⁹ by three or four orders of magnitude, and higher than those of the previous ceramic lithium ionic conductors measured for sintered ceramics, such as Li_3N ($2.4 \times 10^{-4} \text{ S cm}^{-1}$),²⁵ and $\text{La}_{0.62}\text{Li}_{0.14}\text{TiO}_3$ ($1 \times 10^{-4} \text{ S cm}^{-1}$).²⁶ The value is even higher than those of the modified sulfide glasses, $\text{Li}_2\text{S}-\text{SiS}_2$ -LiI ($1.8 \times 10^{-3} \text{ S cm}^{-1}$)¹⁰ and $\text{Li}_2\text{S}-\text{SiS}_2$ - Li_xMO_4 ¹¹⁻¹³ ($\text{Li}_x\text{MO}_4 = \text{Li}_3\text{PO}_4$, Li_4SiO_4 ; 6.9×10^{-4}

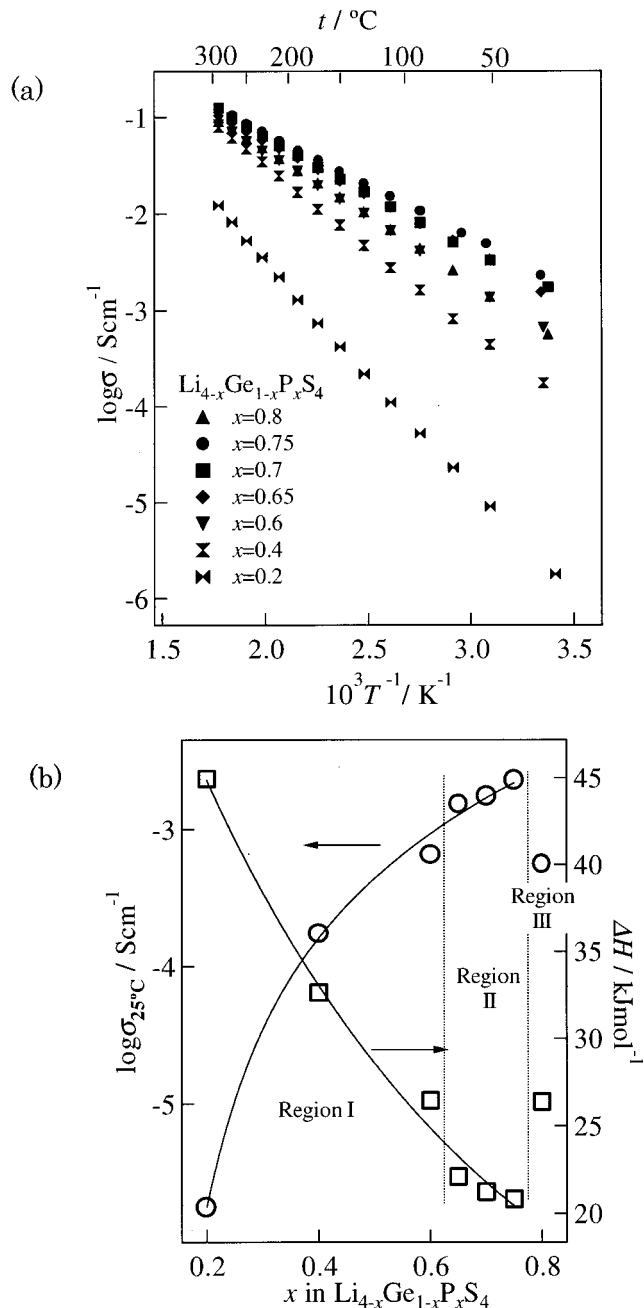


Figure 4. Ionic conductivity data for the new thio-LISICON. (a) Temperature dependence of the conductivity for $\text{Li}_{4-x}\text{Ge}_{1-x}\text{P}_x\text{S}_4$ synthesized at 700°C composition dependence of the conductivities at 25°C (○) and the activation energies (□) for $\text{Li}_{4-x}\text{Ge}_{1-x}\text{P}_x\text{S}_4$. (b) Activation energies were calculated over a temperature range of $25 \leq t \leq 300^\circ\text{C}$.

$- 2 \times 10^{-3} \text{ S cm}^{-1}$); the latter compounds are extensively studied as a candidate of solid electrolyte for all solid lithium battery.¹⁵⁻¹⁷

Chemical and electrochemical stabilities.—Chemical and electrochemical stabilities of ionic conductors are important from a viewpoint of practical applications. The stability the thio-LISICON in contact with lithium was examined.² No reaction was observed with molten lithium up to 200°C . The differential thermal analysis (DTA-TG) measurements of $\text{Li}_{3.25}\text{Ge}_{0.25}\text{P}_{0.75}\text{S}_4$ with lithium metal showed endothermic and exothermic peaks with the same amount of heat anomaly in the heating and cooling process, respectively, at the melting point of lithium ($t \approx 179^\circ\text{C}$); no other reactions were de-

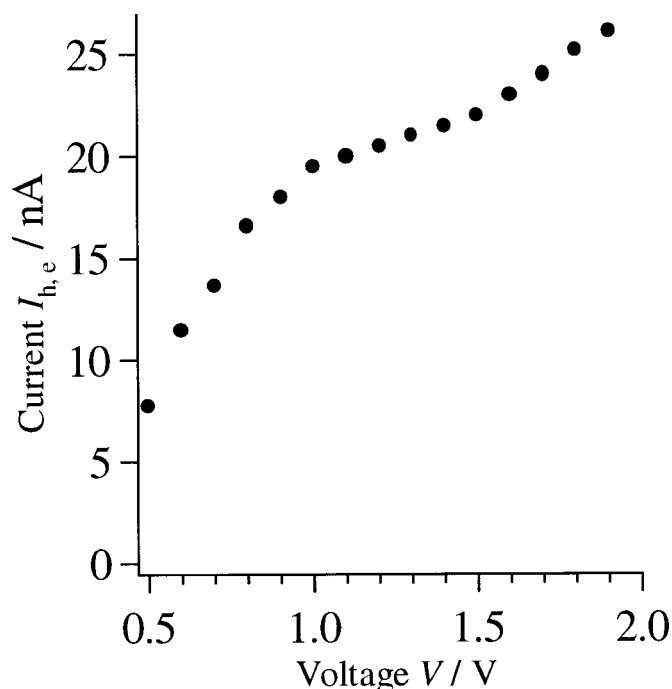


Figure 5. Hebb-Wagner polarization curves obtained from the (–)Li/Li_{3.25}Ge_{0.25}P_{0.75}S₄/Au(+) cell at 25°C.

tested. In addition, the electrochemical stability was examined by CV of the Li/Li_{3.25}Ge_{0.25}P_{0.75}S₄/Au cell using a reference lithium electrode with the scan rate of 1 mV/s and the scan range between –0.5 and 5 V as shown in Fig. 6. Cathodic and anodic currents corresponding to lithium deposition ($\text{Li}^+ + \text{e}^- \rightarrow \text{Li}$) and dissolution ($\text{Li} \rightarrow \text{Li}^+ + \text{e}^-$), respectively, were observed, but no significant currents due to the electrolyte decomposition were detected in the voltage range scanned. Crystalline materials with high ionic conductivity have generally low electrochemical stabilities; *i.e.*, a decomposition potential of 0.44 V for Li₃N²⁷ and a reduction potential of 1.7 V for the La_{1/3–x}Li_{3x}NbO₃ perovskite.²⁸ The thio-LISICON is the first example of crystalline ionic conductors, which have high ionic conductivity and high decomposition potential. Furthermore, the thermal stability, the principal advantage of crystalline materials, was confirmed by the DTA-TG measurement which showed no phase transition over a $25 \leq t \leq 500^\circ\text{C}$ temperature range for Li_{3.25}Ge_{0.25}P_{0.75}S₄.

Material design concepts.—The thio-LISICON family is expressed by a general formula, Li_{4–x}M_{1–y}M'_yS₄ with M = Si, Ge, and M' = P, Al, Zn, Ga. Summarized in Fig. 7 is the material design diagram of the thio-LISICON together with that drawn for the LISICON. The ionic conduction properties are strongly dominated by the size and polarizability of constituent ions, or interstitial vacancy character caused by the substitutions. Note that the conductivities of the thio-LISICON are higher than those of the LISICON. This indicates clearly that the replacement of larger and more polarizable S for O improved conductivity, or larger and more polarizable framework, is preferable for ionic conduction. For the oxides, formation mechanism of the solid solution are well described.⁶ In stoichiometric Li₄SiO₄, for example, the mobile ion concentration is very low and the conductivity is low, whereas in the doped materials, the conductivity rises as the mobile ion concentration (vacancy or interstitials) rises.⁶ Similar to the LISICON, the thio-LISICON is modified based on Li₄SiS₄ and Li₄GeS₄; vacancy-doped systems, Li_{4–x}Si_{1–x}P_xS₄, Li_{4–2x}Zn_xGeS₄, and Li_{4–x}Ge_{1–x}P_xS₄, and interstitial ion-doped systems, Li_{4+x}Si_{1–x}Al_xS₄ and Li_{4+x}Ge_{1–x}Ga_xS₄. When doped to create either Li⁺ vacancies or Li⁺ interstitials, dra-

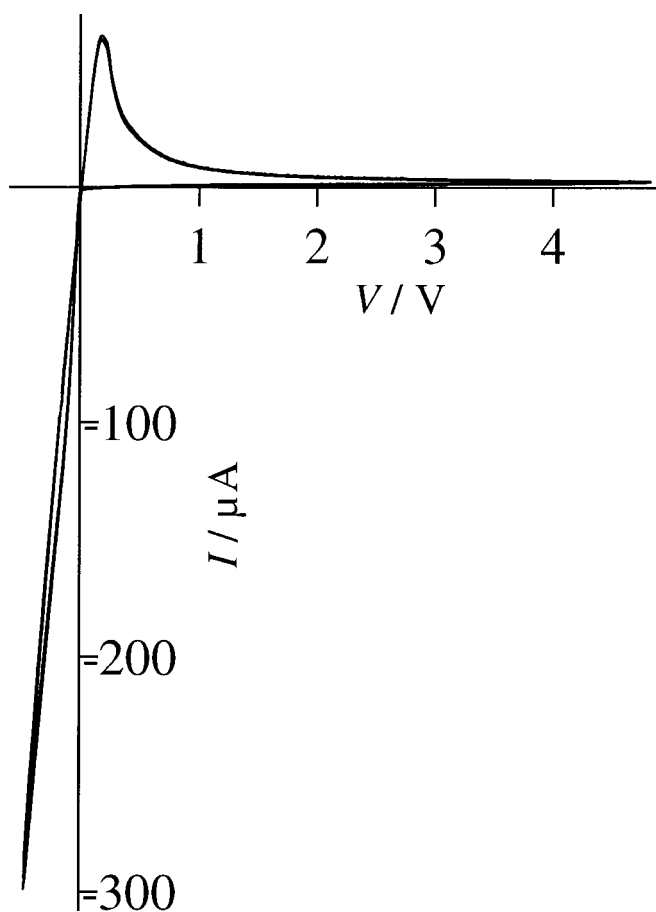


Figure 6. CV of the cell, Li/Li_{3.25}Ge_{0.25}P_{0.75}S₄/Au.

matic increases in conductivity occur: vacancies lead to much higher conductivities than the interstitials do, as can be seen in the diagram. In addition, the germanium system shows higher conductivities than the silicon series, corresponding to its larger volume. Finally, the highest conductivity value was obtained for the Li_{4–x}Ge_{1–x}P_xS₄ system.

Conclusion

Our results show high ionic conductivity in the new thio-LISICON, Li_{4–x}Ge_{1–x}P_xS₄. The material shows negligible electronic conductivity, high electrochemical stability, no reaction with lithium metal, and no phase transition up to 500°C.

Lithium ionic conductors are important as the solid electrolytes of all solid-state battery with free from leaks, available in varied geometries, and high thermal stability, which may improve the safety of the lithium-ion system. A novel approach to form all solid-state batteries is nanocomposites with ceramics and polymers; the glass-polymer composite improved the conductivity and mechanical flexibility of the ceramic electrolytes,²⁹ and the polymer-alumina nanocomposite improved the stability and mechanical strength of lithium polymer electrolytes.³⁰ Attempts to construct energy storage devices for higher operating temperatures will also be available by taking advantage of the thermal stability of the crystalline electrolytes. The thio-LISICON obtained in the present study are members of a material family that a wide variety of ionic conductors belong to, and these materials have potential for applications as bulk ceramic electrolytes and as the conducting components of nanocomposite polymer-based electrolytes.

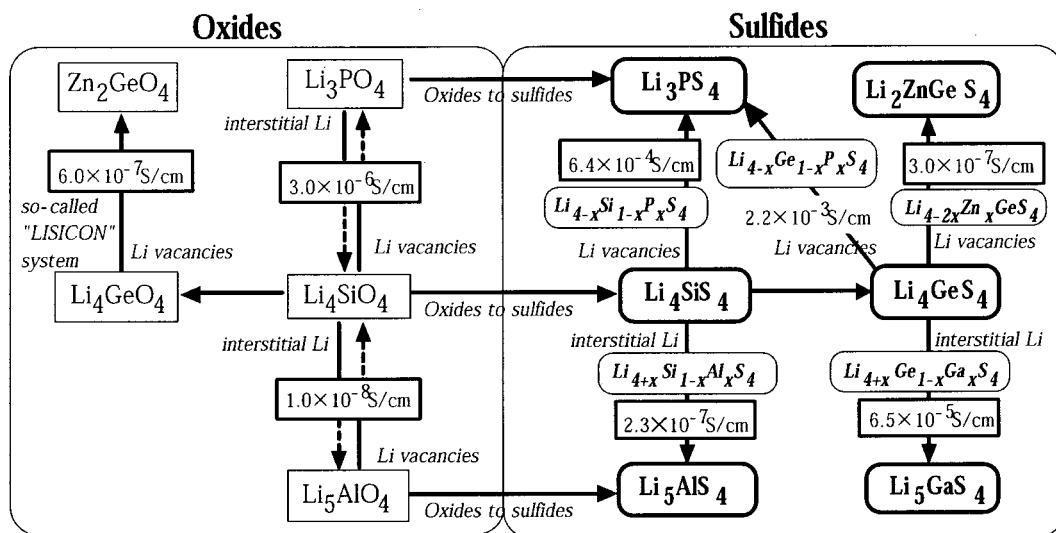


Figure 7. The concept of material design for the LISICON system. Materials belonging to the LISICON (oxides) and the thio-LISICON (sulfides) are summarized. The conductivity data are taken from the systems, $\text{Zn}_2\text{GeO}_4\text{-Li}_4\text{GeO}_4$ (LISICON),⁷ $\text{Li}_3\text{PO}_4\text{-Li}_4\text{SiO}_4$,^{8,9} $\text{Li}_4\text{SiO}_4\text{-Li}_5\text{AlO}_4$,⁸ $\text{Li}_4\text{SiS}_4\text{-Li}_3\text{PS}_4$ (our new results), $\text{Li}_4\text{SiS}_4\text{-Li}_5\text{AlS}_4$ (our new results), $\text{Li}_5\text{GaS}_4\text{-Li}_4\text{GeS}_4$,¹⁸ $\text{Li}_2\text{ZnGeS}_4\text{-Li}_4\text{GeS}_4$,¹⁸ and $\text{Li}_4\text{GeS}_4\text{-Li}_3\text{PS}_4$ (this study).

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